

JURAJ TOMORI

Software Engineer experienced in volumetric video, striving to writing correct, readable, and high-performance code
tomori.juraj@gmail.com, [personal website](#), [learning notes](#)

EXPERIENCE

Dimension Studio

Software Engineer, Applied Technology

May 2024 - Present

Remote

- Implemented a company-wide software distribution in a hybrid environment reducing deployment time from weeks to days: *Intune, Perforce, p4vfs, rez, Python, Chef*
- Developed backend, frontend for our production tooling: *ftrack, Python, Docker, TypeScript, Bootstrap, ECharts*
- Implemented HDR image processing tooling for our capture team: *Python, NumPy, OpenImageIO, LibRaw*

Dimension Studio

Software Developer, R&D

September 2020 - April 2024

Remote

- Implemented motion vectors estimation, colour management for volumetric video which resulted in being awarded multiple productions: *Python, OpenColorIO, FFmpeg, Houdini, Nuke*
- Optimized captured volumetric video data storage by 50%
- Applied differential rendering to volumetric video ([presented at SIGGRAPH 2022](#)): *Python, nvdiffric*
- Implemented distributed post processing pipelines for volumetric video productions: *Whitney Houston: I Wanna Dance with Somebody* ([presented at FMX 2023](#)), *Wicked, Those About to Die*

Dimension Studio

Pipeline & Tools Developer

July 2019 - August 2020

London, England

- Implemented distributed volumetric video postprocessing pipeline reducing average processing time from 5 days to 1 day: *Python, VEX, Houdini, PDG, Natron*
- Implemented renderfarm for efficient processing of large datasets: *HQueue, Ansible, rez*
- Initiated R&D projects focused on improving assets quality for VFX, games, VR, AR and web platforms

Platige Image

FX Artist

July, August 2016, August 2017

Warsaw, Poland

- Short films, game cinematics: *Twardowsky 2.0, Frostpunk, Ghost Recon Wildlands, Skull and Bones, Crossfire HD*

PUBLICATIONS, TEACHING

SIGGRAPH 2021: [Reflectance Estimation for Free-viewpoint Video](#)

SIGGRAPH 2019: [VFX Fractal Toolkit](#)

Filmakademie Baden-Württemberg: [Volumetric video workshop](#), [Programming in VEX](#)

EDUCATION

Filmakademie Baden-Württemberg: *Diploma in Technical Directing*

2016 - 2019

Graduation project: [VFX Fractal Toolkit](#): *Python, OpenCL, OSL, OpenVDB, Houdini, VEX*

Academy of Performing Arts in Bratislava: *B.A. in Visual Effects*

2013 - 2016

Thesis: Crowd simulation in SideFX Houdini: *Python, Houdini, VEX*

COURSES

HarvardX: [CS50's AI with Python](#), [Computer Science](#), [Web development with Python & JavaScript](#)

MITx: [6.00.2x](#), [Computational Thinking with Julia](#), [The Missing Semester of Your CS Education](#)

UW: [Programming Languages](#), **Udacity:** [Interactive 3D Graphics](#)

Dabeaz: [Practical Python Programming](#), [Advanced Python Mastery](#)

TECHNICAL STRENGTHS

Python, NumPy, OpenImageIO, OpenColorIO, FFmpeg, Houdini, TypeScript, ftrack, rez, Perforce, Git, Ansible